

Øving 4 GMPE240 - Simen Røke krogstie

i) Gitt størrelser:

$$H_A = 78.237 \text{ m}$$
$$H_B = 79.582 \text{ m}$$
$$H_C = 75.286 \text{ m}$$

Observasjoner:

$$\begin{array}{ll} \Delta h_{A1} = 1.916 \text{ m} & , D_{A1} = 0.82 \text{ km} \\ \Delta h_{12} = -0.174 \text{ m} & , D_{12} = 1.22 \text{ km} \\ \Delta h_{2C} = -4.691 \text{ m} & , D_{2C} = 1.05 \text{ km} \\ \Delta h_{B1} = 0.560 \text{ m} & , D_{B1} = 0.75 \text{ km} \end{array}$$

Ukjente: H_1 og H_2

Standardavvik:

$$0.002 \text{ m} + 0.002 \text{ m} \cdot D_{\text{km}}$$

ii) Observasjonslikninger:

$$\begin{array}{llll} \Delta h_{A1} = -H_A & + H_1 & \Rightarrow & \Delta h_{A1} + H_A = H_1 \\ \Delta h_{12} = & - H_1 + H_2 & \Rightarrow & \Delta h_{12} = -H_1 + H_2 \\ \Delta h_{2C} = & + H_C - H_2 & \Rightarrow & \Delta h_{2C} - H_C = -H_2 \\ \Delta h_{B1} = & - H_B + H_1 & \Rightarrow & \Delta h_{B1} + H_B = H_1 \end{array}$$

iii) Vektor

$$S_i = 0.002 \text{ m} + 0.002 \text{ m} \cdot D_i, \quad P_i = \frac{S_0^2}{S_i^2}$$

Setter $S_0 = 2.0$

Obs 1: $S_1 = 2.0 \text{ mm} + 2.0 \text{ mm} \cdot 0.82 \text{ km} = 3.64 \text{ mm}$

$$P_1 = \frac{2^2}{3.64^2} = \underline{\underline{0.3019}}$$

Obs 3: $S_3 = 2.0 \text{ mm} + 2.0 \text{ mm} \cdot 1.05 \text{ km} = 4.10 \text{ mm}$

$$P_3 = \frac{2^2}{4.10^2} = \underline{\underline{0.2380}}$$

Obs 2: $S_2 = 2.0 \text{ mm} + 2.0 \text{ mm} \cdot 1.22 \text{ km} = 4.44 \text{ mm}$

$$P_2 = \frac{2^2}{4.44^2} = \underline{\underline{0.3039}}$$

Obs 4: $S_4 = 2.0 \text{ mm} + 2.0 \text{ mm} \cdot 0.75 \text{ km} = 3.5 \text{ mm}$

$$P_4 = \frac{2^2}{3.5^2} = \underline{\underline{0.3265}}$$

iv)

$$A = \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & -1 \\ 1 & 0 \end{bmatrix}, L = \begin{bmatrix} \Delta h_{A1} + H_1 & & & \\ & \Delta h_{A2} & & \\ & \Delta h_{B2} & - & H_C \\ & \Delta h_{B1} & + & H_B \end{bmatrix} = \begin{bmatrix} 80.153 \\ -0.179 \\ -79.977 \\ 80.149 \end{bmatrix}, P = \begin{bmatrix} 0.3619 & 0 & 0 & 0 \\ 0 & 0.2029 & 0 & 0 \\ 0 & 0 & 0.2380 & 0 \\ 0 & 0 & 0 & 0.3265 \end{bmatrix}$$

v)

MKM (Alle beregninger blir gjort med Python)

$$N = A^T P A$$

$$N^{-1} = \begin{bmatrix} 1.3551 & 0.6236 \\ 0.6236 & 2.5551 \end{bmatrix}$$

$$= \begin{bmatrix} 0.8313 & -0.2029 \\ -0.2029 & 0.4409 \end{bmatrix}$$

$$h = A^T P L$$

$$\hat{x} = N^{-1} \cdot h$$

$$= \begin{bmatrix} 50.3999 \\ 18.9993 \end{bmatrix}$$

$$= \begin{bmatrix} 80.1478 \\ 79.9755 \end{bmatrix} = \begin{bmatrix} H_1 \\ H_2 \end{bmatrix}$$

vi) Residualer og feilkvadratsum

$$V = Ax - L$$

$$= \begin{bmatrix} -0.0052 \\ 0.0017 \\ 0.0015 \\ 0.0058 \end{bmatrix}$$

Residualene forteller hvor mye hver obs. må justeres for å passe de estimerte for de ukjente.

$$V^T P V = \underline{\underline{5.0671 \cdot 10^{-4}}}$$

vii) Standardavvik til veksenheten

$$\hat{s}_0 = \sqrt{\frac{V^T P V}{n-2}} = \underline{\underline{0.0032}}$$

Desto lavere standardavviket til veksenheten er, jo mer presis er de estimerte høydene.

viii) $\sigma_{xx} = N^{-1} = \begin{bmatrix} 1.3551 & 0.6236 \\ 0.6236 & 2.5551 \end{bmatrix}$

$$s_{H_1} = s_0 \cdot \sqrt{q_{11}} \\ = 0.0032 \cdot \sqrt{1.3511} \\ = \underline{\underline{0.0037}}$$

$$s_{H_2} = s_0 \cdot \sqrt{q_{22}} \\ = 0.0032 \cdot \sqrt{2.5551} \\ = \underline{\underline{0.0051}}$$

$$i) \Delta h_{12} = \begin{bmatrix} -1 & 1 \end{bmatrix} \begin{bmatrix} H_1 \\ H_2 \end{bmatrix} = -H_1 + H_2 \\ = -80.147 + 79.975 \text{ t} \\ = \underline{\underline{-0.172}}$$

$$C_{22} = \begin{bmatrix} -1 & 1 \end{bmatrix} C_{xx} \begin{bmatrix} -1 \\ 1 \end{bmatrix} \quad (\text{generelle Teilformationslawen})$$

$$C_{xx} = S_0^{-2} \cdot Q_{xx} = \frac{V^T P V}{f} \cdot N^{-1} = 1.024 \cdot 10^{-5} \begin{bmatrix} 1.3511 & 0.6236 \\ 0.6236 & 2.5551 \end{bmatrix} = \begin{bmatrix} 1.281 \cdot 10^{-5} & 6.386 \cdot 10^{-6} \\ 6.386 \cdot 10^{-6} & 2.616 \cdot 10^{-5} \end{bmatrix}$$

$$C_{22} = \begin{bmatrix} -1 & 1 \end{bmatrix} \begin{bmatrix} 1.281 \cdot 10^{-5} & 6.386 \cdot 10^{-6} \\ 6.386 \cdot 10^{-6} & 2.616 \cdot 10^{-5} \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = 2.620 \cdot 10^{-5}$$

$$S_{12} = \sqrt{C_{22}} = \sqrt{2.620 \cdot 10^{-5}} = \underline{\underline{0.00512}}$$